

PIZZERIA PIZZA SALES ANALYSIS

A SQL-driven exploration of a year of pizzeria orders

This report analyzes one year of transactional data (Jan–Dec 2015) from a pizza restaurant using SQL: 21,350 orders, 48,620 line items, 32 pizza types across 4 categories, and 5 sizes. The goal was to answer 13 business questions covering revenue, product performance, order timing, and customer ordering patterns using joins, aggregate functions, subqueries, and window functions.

Key Numbers

21,350 Total Orders	\$817,860 Total Revenue	\$38.31 Avg Order Value
49,574 Pizzas Sold	138.5 Avg Pizzas / Day	32 Pizza Varieties

Methodology

All analysis was performed in MySQL against a four-table relational schema (orders, order_details, pizzas, pizza_types). Techniques used across the 13 queries included:

- Aggregate functions (COUNT, SUM, AVG, ROUND) for volume and revenue metrics
- Multi-table JOINS to connect order line items to pizza attributes and categories
- GROUP BY / ORDER BY / LIMIT for ranking top performers
- Correlated subqueries to compute each pizza type's percentage share of total revenue
- Window functions — SUM() OVER() for running cumulative revenue, and RANK() OVER(PARTITION BY...) to find the top 3 pizzas by revenue within each category

Key Findings

1. Revenue is broad-based, not concentrated

The best-selling pizza (The Thai Chicken Pizza) accounts for only 5.3% of total revenue, and the top 5 pizzas by quantity are within 82 units of each other (2,371–2,453 sold). There is no single "hero product" — revenue is spread evenly across the 32-item menu, which is a sign of a well-balanced menu rather than a single-item dependency.

2. Classic and Chicken pizzas outperform their menu share

Classic pizzas make up 26 of 32 menu items (fewer than Veggie's 27) but generate the most volume (14,888 units, 30% of all pizzas sold). Chicken pizzas, despite having the fewest varieties on the menu (18), rank #1 in revenue-per-item — 3 of the top 5 highest-revenue pizzas are Chicken category. This suggests Chicken pizzas are priced and/or positioned more effectively than their catalog size would predict.

Category	Units Sold	Top Revenue Pizza
Classic	14,888	The Classic Deluxe Pizza
Supreme	11,987	The Spicy Italian Pizza
Veggie	11,649	The Four Cheese Pizza
Chicken	11,050	The Thai Chicken Pizza

3. Large pizzas dominate, but the premium sizes are nearly untouched

Large (L) is the most-ordered size at 18,956 units (38%), followed by Medium and Small. XL and XXL combined make up just 1.2% of all pizzas sold (580 of 49,574 units) — despite XXL carrying the highest price point on the menu (\$35.95 for The Greek Pizza). This is the clearest underperformer in the dataset and the easiest lever to test.

4. Ordering is heavily concentrated around two daily peaks

Orders cluster sharply around lunch (12–1 PM, ~2,500 orders/hour) and dinner (5–6 PM, ~2,300–2,400 orders/hour), with a visible mid-afternoon lull (2–3 PM). Virtually no orders occur before 11 AM or after 11 PM. Staffing and prep schedules that mirror this curve — rather than a flat shift pattern — would directly reduce idle labor cost during the 2–4 PM and post-9 PM windows.

5. Revenue growth held steady, without seasonal spikes

Cumulative revenue grew at a highly consistent linear rate across all 358 active days of 2015, averaging roughly \$2,284 in daily revenue with no major seasonal dips or spikes. This indicates stable, predictable demand — useful for inventory and staffing forecasts, but also a signal that no current marketing or seasonal promotion is meaningfully moving the needle on order volume.

Recommendations

- Test a price or bundling change on XL/XXL sizes, or consider trimming them — they contribute negligible volume relative to shelf space and menu complexity.
- Align staffing schedules to the two-peak demand curve (12–1 PM and 5–6 PM) instead of flat shifts, to cut idle labor during the 2–4 PM trough.
- Investigate what makes Chicken-category pizzas over-perform their catalog size — pricing, ingredients, or menu placement — and apply those learnings to Veggie, which has the most menu items but the lowest revenue.
- Since no seasonal spike exists in the data, a scheduled promotion (e.g., a slow-Tuesday discount) is untested territory with clear upside potential.

Business Questions - Answered

#	Questions	Answers
1	Total number of orders placed	21,350 orders
2	Total revenue generated from pizza sales	\$817,860.05
3	Highest-priced pizza	The Greek Pizza - \$35.95
4	Most common pizza size ordered	Large (L) - 18,956 orders
5	Top 5 most ordered pizza types by quantity	• The Classic Deluxe Pizza — 2,453 • The Barbecue Chicken Pizza — 2,432 • The Hawaiian Pizza — 2,422 • The Pepperoni Pizza — 2,418 • The Thai Chicken Pizza — 2,371
6	Total quantity ordered per pizza category	• Classic — 14,888 • Supreme — 11,987 • Veggie — 11,649 • Chicken — 11,050
7	Distribution of orders by hour of day	11 AM — 1,231 12 PM — 2,520 1 PM — 2,455 2 PM — 1,472 3 PM — 1,468 4 PM — 1,920 5 PM — 2,336 6 PM — 2,399 7 PM — 2,009 8 PM — 1,642 9 PM — 1,198 10 PM — 663 11 PM — 28
8	Category-wise distribution of pizza varieties	• Classic — 9 pizza types • Supreme — 9 pizza types • Veggie — 9 pizza types • Chicken — 6 pizza types
9	Average number of pizzas ordered per day	Approximately 138 pizzas/day
10	Top 3 pizza types by revenue	• The Thai Chicken Pizza — \$43,434.25 • The Barbecue Chicken Pizza — \$42,768.00 • The California Chicken Pizza — \$41,409.50
11	Revenue contribution % of each pizza type	• Classic — 26.91% • Supreme — 25.46% • Chicken — 23.96% • Veggie — 23.68%
12	Cumulative revenue over time	\$817,860.05
13	Top 3 pizza types by revenue, within each category	• Chicken

#	Questions	Answers
		• The Thai Chicken Pizza • The Barbecue Chicken Pizza • The California Chicken Pizza • Classic • The Classic Deluxe Pizza • The Hawaiian Pizza • The Pepperoni Pizza • Supreme • The Spicy Italian Pizza • The Italian Supreme Pizza • The Sicilian Pizza • Veggie • The Four Cheese Pizza • The Mexicana Pizza • The Five Cheese Pizza